

Basics of Artificial Neural Networks (ANN): Supervised, Unsupervised and Reinforcement Learning.

6. Department of Delhi School of Management

Managerial themes such as E-Governance, Information Technology Management, Strategic Management, Marketing Management, Distribution and Retail Management, Organizational Behavior, Human Resources Management, Corporate Governance and Ethics, Public Policy and Governance, Accounting and Finance, Portfolio Management, Mergers and Acquisition,

Corporate Restructuring, Knowledge Management, General Management Principle and Practices, Supply Chain Management, Business Research Methods, Business Statistics.

7. Department of Design

1. Visualization and Spatial ability
2. Environmental and Social Awareness
3. Analytical and Logical Reasoning
4. Language and Creativity
5. Observation and Design Sensitivity

8. Department of Electrical Engineering/ COE-EVRT

Section 1 : Electric Circuits

Network graph, KCL, KVL, Node and Mesh analysis, Transient response of dc and ac networks, sinusoidal steady-state analysis, resonance, passive filters, Ideal current and voltage sources, Thevenin's theorem, Norton's theorem, Superposition theorem, Maximum power transfer theorem, Two-port networks, Three phase circuits, Power and power factor in ac circuits.

Section 2 : Electromagnetic Fields

Coulomb's Law, Electric Field Intensity, Electric Flux Density, Gauss's Law, Divergence, Electric field and potential due to point, line plane and spherical charge distributions, Effect of dielectric medium, Capacitance of simple configurations, Biot-Savart's law, Ampere's law,

Curl, Faraday's law, Lorentz force, Inductance, Magneto motive force, Reluctance, Magnetic circuits, Self and Mutual inductance of simple configurations.

Section 3 : Signals and Systems

Representation of continuous and discrete-time signals, Shifting and scaling operations, Linear Time Invariant and Causal systems, Fourier series representation of continuous periodic signals, sampling theorem, Applications of Fourier Transform, Laplace Transform and z-Transform.

Section 4 : Electrical Machines

Single phase transformer; equivalent circuit, phasor diagram, open circuit and short circuit tests, regulation and efficiency; Three phase transformer; connections, parallel operation; Auto-transformer, Electron mechanical energy conversion principles, DC machines;

separately excited, series and shunt, motoring and generating mode of operation and their characteristics, starting and speed control of dc motors; Three phase induction motors, Principle of operation, types, performance, torque-speed characteristics, no-load and blocked rotor tests, equivalent circuit, starting and speed control; Operating principle of single phase induction motors; Synchronous machines: cylindrical and salient pole machines, performance, regulation and parallel operation of generators, starting of synchronous motor, characteristics; Types of losses and efficiency calculations of electric machines.

Section 5 : Power systems

Power generation concepts, ac and dc transmission concepts, Models and performance of transmission lines and cables, Series and shunt compensation, Electric field distribution and insulators, Distribution systems, Per-unit quantities, Bus admittance matrix, Gauss-Seidel and Newton-Raphson load flow methods, Voltage and Frequency control, power factor correction, Symmetrical components, Symmetrical and unsymmetrical fault analysis, Principles of over-current, differential and distance protection; Circuit breakers, System stability concepts, Equal area criterion.

Section 6 : Control Systems

Mathematical modeling and representation of systems, Feedback principle, transfer function, Block diagrams and Signals flow graphs, Transient and Steady-state analysis of linear time invariant systems, Routh-Hurwitz and Nyquist criteria, Bode plots Root loci, Stability analysis, Lag, Lead and Lead-Lag compensators; P, PL and PID controllers; State space model, State transition matrix.

Section 7 : Electrical and Electronic Measurements

Bridges and potentiometers, Measurement of voltage, current, power, energy and power factor; Instrument transformers, Digital voltmeters and multimeters, Phase, Time and Frequency measurement; Oscilloscopes, Error analysis.

Section 8 : Analog and Digital Electronics

Characteristics of diodes, BJT, MOSFET; simple diode circuits clipping, clamping, rectifiers; Amplifiers; Biasing, Equivalent circuit and Frequency response; Operational amplifiers; Characteristics and applications, Combinational and Sequential logic circuits, multiplexer, DE multiplexer, 8085 microprocessors; Architecture, programming and Interfacing.

Section 9 : Power Electronics

Characteristics of semiconductor power devices; Diode, Thyristor, Triac, GTO, MOSFET, IGBT; DC to DC conversion; Buck, Boost and Buck-Boost converters; Single and three phase configuration of uncontrolled rectifiers, Line commutated thyristor based converters, Bidirectional ac to dc voltage source converters, Issues of line current harmonics, Power factor, Distortion factor of ac to dc converters, Single phase and three phase inverters, Sinusoidal pulse width modulation.

9. Department of Electronics and Communication Engineering

Networks, Signals and Systems

Network Analysis: Node and mesh Analysis, superposition, Thevenin's theorem and Norton's theorem, maximum power transfer, Steady state sinusoidal analysis: phasors;

Time and frequency domain analysis of linear circuits; Solution of network equations using Laplace transform; Frequency domain analysis of RLC circuits.

Continuous-Time Signals: Fourier series and Fourier transform, sampling theorem and applications.

Discrete-Time Signals: DTFT, DFT, FFT, Z-transform, interpolation of discrete-time signals (FIR, IIR).

LTI systems; definition and properties causality, stability, impulse response, convolution, poles and zeros, parallel and cascade structure, frequency response, group delay, phase delay, digital filter design techniques.

Image Processing

Image Transforms: Unitary transforms, Sine, Cosine, Hadamard transform, KL transform, Short term Fourier transform, Wavelet transform, DWT.

Image Enhancement Techniques- Histogram Modelling, Spatial operations, transforms operations.

Frequency Domain- smoothing frequency domain filters-sharpening frequency domain filters, homomorphic filtering.

Segmentation Techniques- Thresholding based, cluster analysis, region growing.

Morphological Operation- Dilation, Erosion. Histogram and Histogram equalization. Feature Extraction Techniques. Image restoration-Weiner filter, Image reconstruction radon transform and inverse radon transform.

Electronic Devices

Energy bands in intrinsic and extrinsic semiconductors; Carrier transport; diffusion current, drift current, mobility and resistivity; Generation and recombination of carriers; Poisson and continuity equations; P-N junction, Zener diode, BJT, MOS capacitor, MOSFET, LED, photodiode and solar cell; Integrated circuit fabrication process; oxidation, diffusion, ion implantation, photo lithography, and twin-tub CMOS process.